

Subject: Engineering Physics

Code: BTE22010

4 Credits | Semester II

A. Introduction:

- To acquire fundamental knowledge about nature and its phenomena including quantitative expression.
- To enhance intellectual, computational, experimental, communication and analytical skills of the students Physics is necessary to satisfy the basic sciences requirement, as appropriate for various engineering disciplines

B. Course Outcomes: At the end of the course, students will be able to

[CO1] Identify and understand the kinds of experimental results which are incompatible with classical Physics leading to the development of a quantum theory of matter and light.

[CO2] Use basic concepts to analyze and design a wide range of semiconductor devices.

[CO3] Understand & solve different types of wave equations.

[CO4] Use the principles of optics to solve various complex engineering problems.

[CO5] Use fundamental laws and relations to solve problems in electricity, electromagnetism

C. Assessment Plan:

Criteria	Description	Maximum Marks
Continuous Internal Assessment (CIA)	Internal Examination	20
	Attendance	5
	Assignment	5
End Semester Examination(ESE)	End Semester Examination	70
Total		100
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking up the End Semester examination. The allowance of 25% includes all types of leaves including medical leaves.	

D. SYLLABUS

QUANTUM MECHANICS: Introduction to Quantum Physics, black body radiation , Explanation using the photon concept, Photoelectric effect, Compton effect, Wave particle duality, De-Broglie hypothesis, Heisenberg's Uncertainty principle. Born interpretation for wave function, Free-particle wave function and wave-packets, Time-dependent and time independent Schrodinger equation, particle in a box, Finite Potential barrier and tunneling.

ELECTRONIC MATERIALS AND SEMICONDUCTOR: Free electron theory, Energy bands in solids, E-k diagram, Direct and indirect band gaps, Types of electronic materials: metals, semiconductors, and insulators, Density of states, Occupation probability, Fermi level, and Effective mass. Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier-concentration and temperature, Carrier generation and recombination, Carrier transport: diffusion and drift, p-n junction,

WAVES AND OSCILLATION: Simple harmonic motion, damped and forced simple harmonic oscillator Mechanical and electrical simple harmonic oscillators, damped harmonic oscillator – heavy,

critical and light damping, energy decay in a damped harmonic oscillator, quality factor, forced mechanical and electrical oscillators, electrical and mechanical impedance, resonance.

OPTICS AND LASER: Introduction to interference, Analytical treatment of interference, Displacement of fringes, Thin film, Wedge shaped film, Newton's Ring, Concept of diffraction, Fraunhofer and Fresnel diffraction, Fraunhofer diffraction at single slit, double slit, and diffraction grating, Limit of Resolution, Resolving power of grating, Introduction to interaction of radiation with matter, Stimulated and spontaneous emission, Einstein's coefficient, principles and working of laser: population inversion, pumping, types of laser: He-Ne laser, Ruby laser, application of lasers.

ELECTROMAGNETISM: Electric field and electrostatic potential for a charge distribution; Laplace's and Poisson's equations for electrostatic potential and uniqueness of their solution. Boundary conditions of electric field and electrostatic potential, Bio-Savart law, vector potential and calculating it for a given magnetic field; the equation for the vector potential and its solution for given current densities. Faraday's law of electromagnetic induction, Equation of continuity, displacement current, Maxwell's equation, Poynting theorem, Electromagnetic waves in free space, conducting and non-conducting medium

E. TEXT BOOKS

- T1. Concept of Modern Physics by Arthur Beiser: Publication: TMH
- T2. Elements of electro magnetics by Mathew N.O. Sadiku: Publication: Oxford University Press
- T3. Introduction to electrodynamics by David J. Griffiths; Pub.: Pearson Education.
- T4. Optics by Ajoy Ghatak Pub; TMH
- T5. Fundamentals of Physics extended volume by Resnick, Halliday and Walker; Pub.: John Wiley & Sons. Inc. Asian Edition.
- T6. P. Bhattacharya, Semiconductor Optoelectronic Devices, Prentice Hall of India (1997).
- T7. Engineering Physics – Hitendra K. Malik & Ajay Kumar Singh by TMH Publication.
- T8. The Physics of Vibrations and Waves, H. J. Pain, 2013, John Wiley and Sons.
- T9. Introduction to Quantum mechanics, Nikhil Ranjan Roy, 2016, Vikash Publishing House Pvt Ltd

F. REFERENCE BOOKS

- R1. Modern Physics by G. Aruldas & P. Rajagopal; Pub: Prentice Hall of India.
- R2. Quantum Physics by H.C. Verma Pub.: Surya Publication .
- R3. Lasers and Non-Linear Optics by B.B. Laud; Pub: New Age International (P) Ltd.
- R4. Principles of electricity by Leigh Page and Norman Ilesley Adams, Pub.: Eurasia Publishing House, New Delhi.
- R5. Feynmann Lecture Series on Physics
- R6. Waves: Berkeley Physics Course, vol.3, Francis Crawford, 2007, Tata McGraw-Hill.
- R7. Online course: "Semiconductor Optoelectronics" by M R Shenoy on NPTEL
- R8. Introduction to Modern Physics, Rich Meyer, Kennard, Coop, 2002, Tata McGraw Hill